INDIAN INSTITUTE OF INFORMATION TECHNOLOGY TIRUCHIRAPALLI

Department of Computer Science and Engineering

WildSnake

Snake Game Engine: Object-Oriented Design and Implementation

Mini Project Report

Course: CS3104 – Object-Oriented Programming in C++

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Final project report

Declaration

We declare that the project contribution described in this report, titled *WildSnake*, was developed as a CS3104 mini project. The implementation details and limitations are reported to the best of our knowledge. Any external libraries used are identified in the report. The student authors are listed on the cover page.

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Acknowledgement

We thank the course instructor for the instruction and feedback provided during this mini project. (Edit this section to acknowledge any additional contributors.)

Abstract

WildSnake is a small snake-game engine implemented in C++17 and presented through a browser interface. The engine models the snake, food, board, score, and game state as separate C++ classes. A lightweight HTTP server exposes state, direction, tick, and reset operations to a JavaScript front end. The project demonstrates encapsulation, composition, an abstract observer interface, standard-library containers, and separation between game rules and presentation. The repository also contains a CMake build and a set of assertion-style tests. The report documents the current source structure and does not claim that the project has been compiled or that the tests have passed.

Keywords: C++, object-oriented programming, game engine, Snake, REST API, UML, CMake

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Chapter 1

Introduction

1.1 Background

Snake is a grid-based game in which the player steers a growing snake to collect food while avoiding the board boundary and the snake's own body. Its compact rules make it useful for demonstrating state, object collaboration, input validation, collision checks, and separation of responsibilities.

1.2 Project overview

WildSnake separates the rule engine from the graphical interface. The C++ backend owns the authoritative game state. An HTTP API lets the front end read the state and request direction changes, game ticks, and resets. The interface uses HTML, CSS, and JavaScript with a canvas for the board.

1.3 Problem statement

The project aims to provide a maintainable, small-scale implementation of the classic Snake rules, with a clear object-oriented model and a usable interface. The design should make core entities and their responsibilities understandable to a student reviewing OOP concepts.

1.4 Objectives

- Model game entities with focused C++ classes and private state.
- Implement movement, direction validation, food placement, scoring, and collision/game-over behavior.
- Expose game operations through a small JSON HTTP API.

- Provide keyboard, on-screen, start, pause, and restart controls.
- Document the architecture and test the core behavior.

Chapter 2

Scope and Requirements

2.1 Scope

The current implementation has a rectangular, non-wrapping board, a snake starting at the board center, randomly placed food, score increments when food is eaten, and game-over detection for wall, self, and obstacle collisions. The world contains an obstacle collection, although the current constructor does not populate it. The browser UI supports arrow keys and WASD, clickable direction buttons, pause/resume, and restart. The backend listens on `127.0.0.1:8080`.

The score manager can load and save a high score in a text file, but it is not connected to the `Game` score lifecycle. The browser's high-score presentation therefore should not be described as integrated persistent scoring without further implementation and verification.

2.2 Functional requirements

Table 2.1: Functional requirements reflected in the current source

ID	Requirement
FR-01	Display current game state, including board dimensions, snake, food, obstacles, score, and game-over flag.
FR-02	Accept one of four cardinal direction requests.
FR-03	Advance the snake one cell per tick and check collisions.
FR-04	Increase the score and place new food when the snake reaches food.
FR-05	Reset the snake and score and resume active play.
FR-06	Allow browser controls by keyboard and on-screen buttons.
FR-07	Support pause/resume in the front-end game loop.
FR-08	Load and save a high score through the standalone ScoreManager class.

2.3 Non-functional requirements

The code targets C++17 and uses CMake. The engine should keep game rules independent from browser rendering, validate board dimensions and directions, and provide readable class interfaces. The server uses a mutex around shared game operations. Third-party dependencies declared by CMake are cpp-httplib and nlohmann-json.

2.4 Use cases

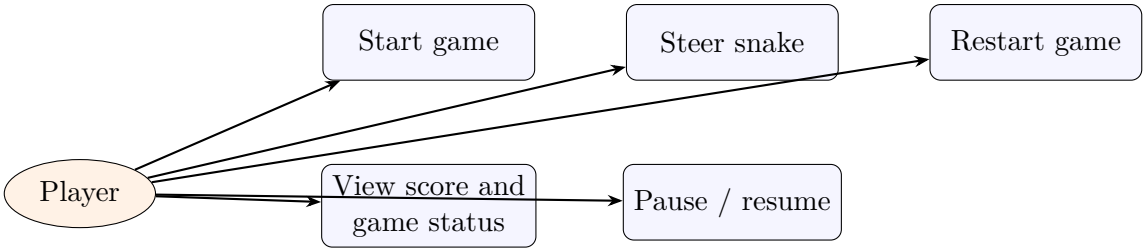


Figure 2.1: Use-case view of player interactions

Chapter 3

Design and Architecture

3.1 Layered architecture

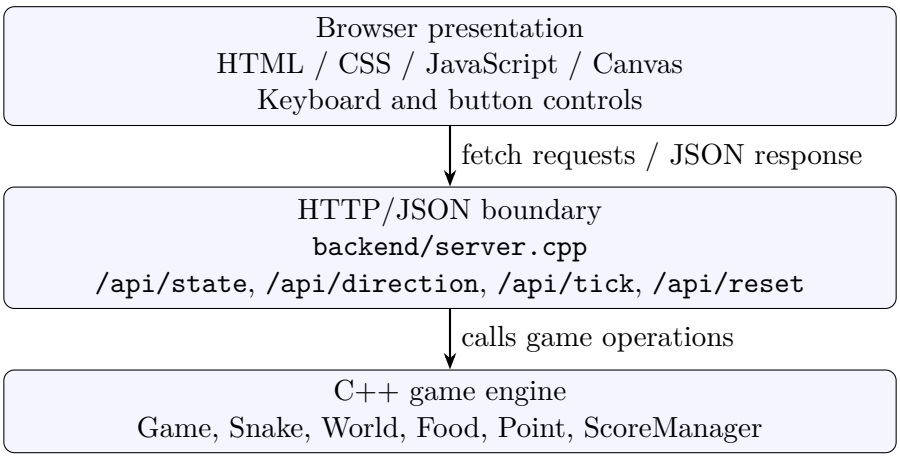


Figure 3.1: WildSnake high-level architecture

3.2 Class design and OOP concepts

- **Encapsulation:** game state is held in private members and read through public methods such as `getScore()` and `getBody()`.
- **Composition:** `Game` owns a `Snake` and a `World`; `World` owns a `Food` value.
- **Abstraction and runtime polymorphism:** `GameObserver` declares virtual event callbacks, and `Game` calls them through the interface. A concrete observer can be supplied by a client.
- **Standard library:** `std::deque<Point>` represents the body, vectors represent obstacles and occupied cells, and `std::mt19937` supports randomized food placement.
- **Value type:** `Point` represents a coordinate and defines equality operators used by collision and collection logic.

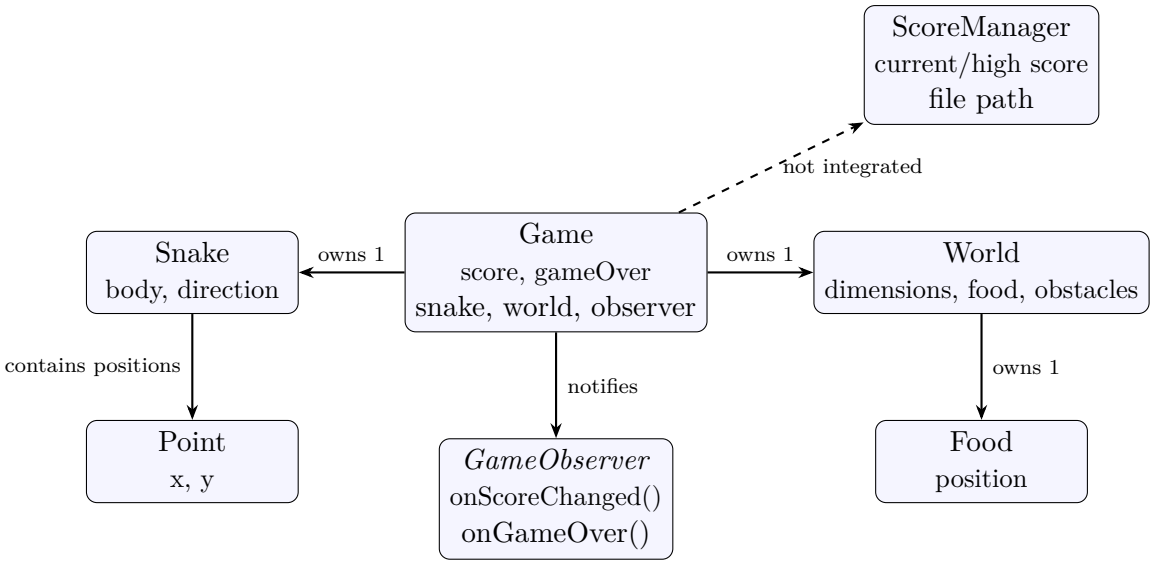


Figure 3.2: Core class relationships. ScoreManager is currently standalone.

3.3 UML class specification tables

The following UML-style tables list the declared attributes and public operations for every class in the game model. A minus sign denotes private visibility and a plus sign denotes public visibility. The `GameObserver` callbacks are virtual; `ScoreManager` remains a separate utility in the current implementation.

Table 3.1: UML class table: model data types and entities

Class	Attributes (visibility)	Public operations (visibility)
`Point`	`x: int`, `y: int` (public value fields)	`Point(x=0,y=0)`; `operator==`; `operator!=`.
`Direction` (enum)	Values: `UP`, `DOWN`, `LEFT`, `RIGHT`	Scoped enumeration used by `Snake` and the API.
`Snake`	`body_: deque<Point>`, `direction_: Direction`, `growing_: bool` (private)	`Snake(start)`; `getBody()`; `getDirection()`; `setDirection()`; `move()`; `grow()`; `occupies()`; `hitsItself()`.
`Food`	`position_: Point` (private)	`Food(position)`; `getPosition()`; `setPosition()`.
`Grid<T>`	`width_: int`, `height_: int`, `cells_: vector<T>` (private)	`Grid(width,height)`; `getWidth()`; `getHeight()`; `contains(Point)`; `at(Point)` (mutable and const overloads). Methods are declared in the header; definitions are absent from the supplied source.

Class	Attributes (visibility)	Public operations (visibility)
`World`	`width_: int`, `height_: int`, `food_: Food`, `obstacles_: vector<Point>`, `rng_: mt19937` (private)	`World(width,height)`; `getWidth()`; `getHeight()`; `getFood()`; `getObstacles()`; `isInside(Point)`; `hasObstacle(Point)`; `placeFood(occupied)`.

Table 3.2: UML class table: game coordination and score utility

Class	Attributes (visibility)	Public operations (visibility)
`GameObserver` (abstract interface)	No data members declared	`onScoreChanged(score)` (public pure virtual); `onGameOver()` (public pure virtual); virtual destructor.
`Game`	`snake_: Snake`, `world_: World`, `score_: int`, `gameOver_: bool`, `observer_: unique_ptr<GameObserver>` (private)	`Game(width,height)`; `setObserver()`; `changeDirection()`; `tick()`; `reset()`; `getScore()`; `isGameOver()`; `getSnake()`; `getWorld()`.
`ScoreManager`	`currentScore_: int`, `highScore_: int`, `filePath_: string` (private)	`ScoreManager(filePath)`; `resetCurrentScore()`; `addPoints(points)`; `getCurrentScore()`; `getHighScore()`; `loadHighScore()`; `saveHighScore() const`. This utility is not currently composed into `Game`.

3.4 Sequence: one game tick

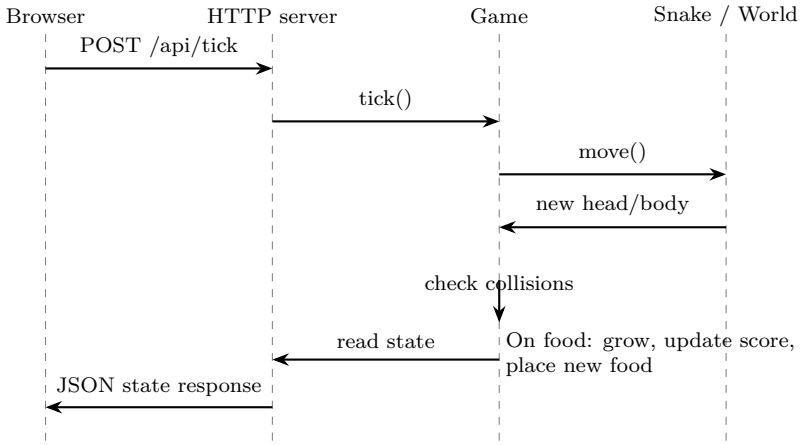


Figure 3.3: Sequence for a browser-requested game tick

3.5 Activity: tick and collision handling

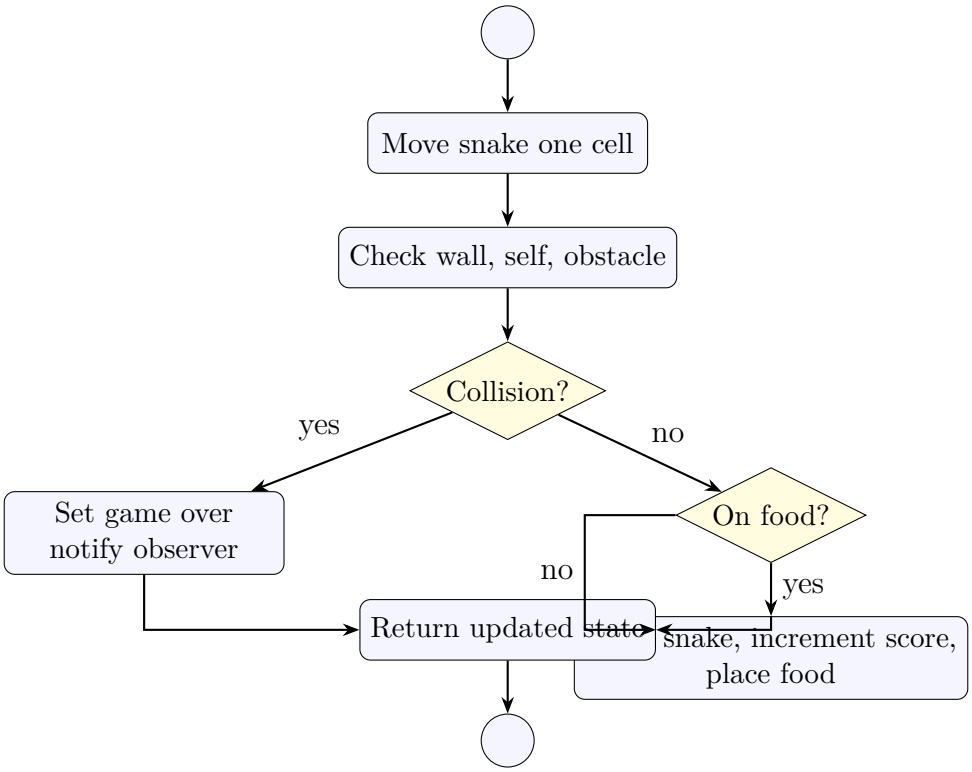


Figure 3.4: Activity diagram for a game tick

Chapter 4

Implementation

4.1 Technology stack

WildSnake uses a lightweight browser frontend and a C++17 backend. The frontend is implemented with native browser technologies and does not use a JavaScript framework. The C++ game engine owns the authoritative gameplay state; an HTTP/JSON interface carries input requests to the engine and returns state for the browser to render.

Table 4.1: Technology stack and its role in WildSnake

Technology	Role in the project
C++17 and standard library	Implement the game engine and model. The project uses containers such as `deque` and `vector`, random-number utilities, smart pointers, file streams, and a mutex.
HTML5 and CSS3	Structure and style the browser interface.
JavaScript and Canvas	Handle keyboard and button input, call the backend, run the browser game loop, and draw the board and game entities.
HTTP and JSON	Connect the frontend and backend through the state, direction, tick, and reset endpoints.
cpp-httplib	Provide the lightweight C++ HTTP server; CMake fetches version 0.18.3.
nlohmann-json	Parse and serialize JSON in the server; CMake fetches version 3.12.0.
CMake and CTest	Configure the C++17 build and register the game test executable. The presence of the test target does not by itself confirm that the tests pass.
Python 3	Serve the static frontend locally during development using Python's HTTP server.
Browser local storage	Store the high-score value shown by the frontend; this is separate from the standalone C++ `ScoreManager`.
Git and GitHub	Track source changes and host the project repository.

The backend dependencies are declared in `backend/CMakeLists.txt` and are fetched by CMake. The project uses no frontend framework: the browser modules are plain JavaScript files that communicate with the local C++ server at `127.0.0.1:8080`.

4.1.1 Project modules

Table 4.2: Project modules

Module	Responsibility
`Snake`	Body, direction validation, movement, growth, self-collision.
`World` and `Food`	Bounds, obstacles, food position, randomized placement.
`Game`	Coordinates ticks, collisions, score, reset, observer notifications.
`ScoreManager`	Reads/writes persistent high score in a text file.
`server.cpp`	Serializes state as JSON and routes HTTP operations.
Frontend	Canvas rendering, controls, API calls, and game-loop presentation.
Tests	Small executable covering points, snake, world, grid, and basic game state.

4.2 Representative OOP code

The following excerpts are drawn from the current source with formatting normalized for the report.

4.2.1 Encapsulated game model and observer abstraction

Listing 4.1: Game owns collaborators and exposes a narrow interface

```
class GameObserver {
public:
    virtual void onScoreChanged(int score) = 0;
    virtual void onGameOver() = 0;
    virtual ~GameObserver() = default;
};

class Game {
private:
    Snake snake_;
    World world_;
    int score_;
    bool gameOver_;
    std::unique_ptr<GameObserver> observer_;
public:
    void tick();
    void reset();
    int getScore() const;
    bool isGameOver() const;
```

```
};
```

4.2.2 Direction validation and movement

Listing 4.2: Snake validates directions and moves its deque body

```
void Snake::setDirection(Direction direction)
{
    switch (direction) {
        case Direction::UP: case Direction::DOWN:
        case Direction::LEFT: case Direction::RIGHT:
            break;
        default:
            throw std::invalid_argument("Invalid snake direction");
    }
    // A multi-segment snake cannot reverse into itself.
    if (body_.size() > 1 && isOpposite(direction_, direction))
        return;
    direction_ = direction;
}

void Snake::move()
{
    Point head = body_.front();
    advance(head, direction_);
    body_.push_front(head);
    if (growing_) growing_ = false;
    else body_.pop_back();
}
```

Note: `isOpposite` and `advance` above are explanatory helpers; the repository implements these checks directly in the method body.

4.2.3 Tick orchestration and composition

Listing 4.3: Game coordinates movement, collisions, and food handling

```
void Game::tick()
{
    if (gameOver_) return;
    snake_.move();
    checkCollisions();
    if (gameOver_) return;

    if (snake_.getBody().front() == world_.getFood().getPosition()) {
        snake_.grow();
        updateScore();
        // Build occupied cells and request the next food position.
        world_.placeFood(occupiedCells());
```

```
    }
}
```

Note: `occupiedCells()` abbreviates the vector construction in the source; the repository builds that vector from the snake body.

4.3 HTTP API

Table 4.3: Backend API operations

Method	Path	Behavior
GET	`/api/state`	Returns dimensions, snake, food, obstacles, score, gameOver.
POST	`/api/direction`	Accepts JSON `{"direction":"UP"}`; rejects invalid input.
POST	`/api/tick`	Advances one step and returns updated state.
POST	`/api/reset`	Resets the engine and returns new state.
GET	`/`	Returns server name/status and endpoint names.

Chapter 5

Testing and Results

5.1 Test approach

The repository contains `tests/game_tests.cpp`, a standalone test program registered with CTest when `BUILD_TESTING` is enabled. Its checks cover point equality, initial and changed snake movement, invalid direction rejection, world boundaries and minimum size, grid access, initial game state, one tick, and reset. The test file expresses checks by printing PASS/FAIL and returning a failing exit code if any check fails.

5.2 Test status and observed repository limitations

The report was prepared by source inspection; no build or test execution is claimed. Before submission, compile the project and record the actual command, environment, and result here. At source inspection, there are case/name inconsistencies that may prevent a clean build on case-sensitive systems: `point.h` versus includes of `Point.h`, and `ScoreMananger.h` versus includes of `ScoreManager.h`. Also, `Grid<T>` declares template methods in `World.h`, but their definitions are not present in the supplied source files even though the tests instantiate `Grid<int>`. Resolve these and rerun the tests before submitting.

Table 5.1: Coverage in the existing test source (not a claim of passing tests)

Area	Checks represented
Point	Equality and inequality.
Snake	Initial position, movement, direction change, self-hit baseline, invalid direction.
World and Grid	In-bounds checks, minimum dimensions, element access and out-of-range behavior.
Game	Initial score/state, one tick, score/state after reset.
Additional recommended cases	Eat/grow/score, wall collision, self-collision, reverse-direction prevention, no free food cells, observer callbacks, server request validation.

5.3 Screenshots

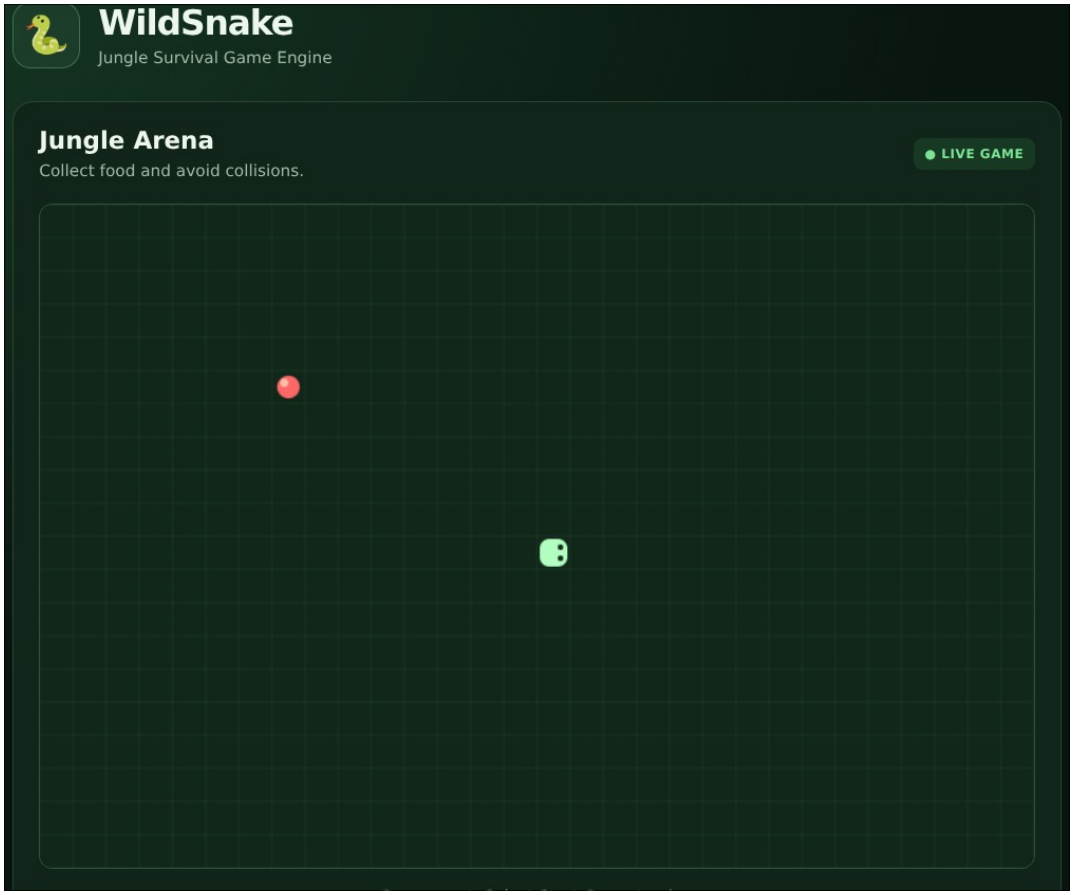


Figure 5.1: WildSnake initial game screen

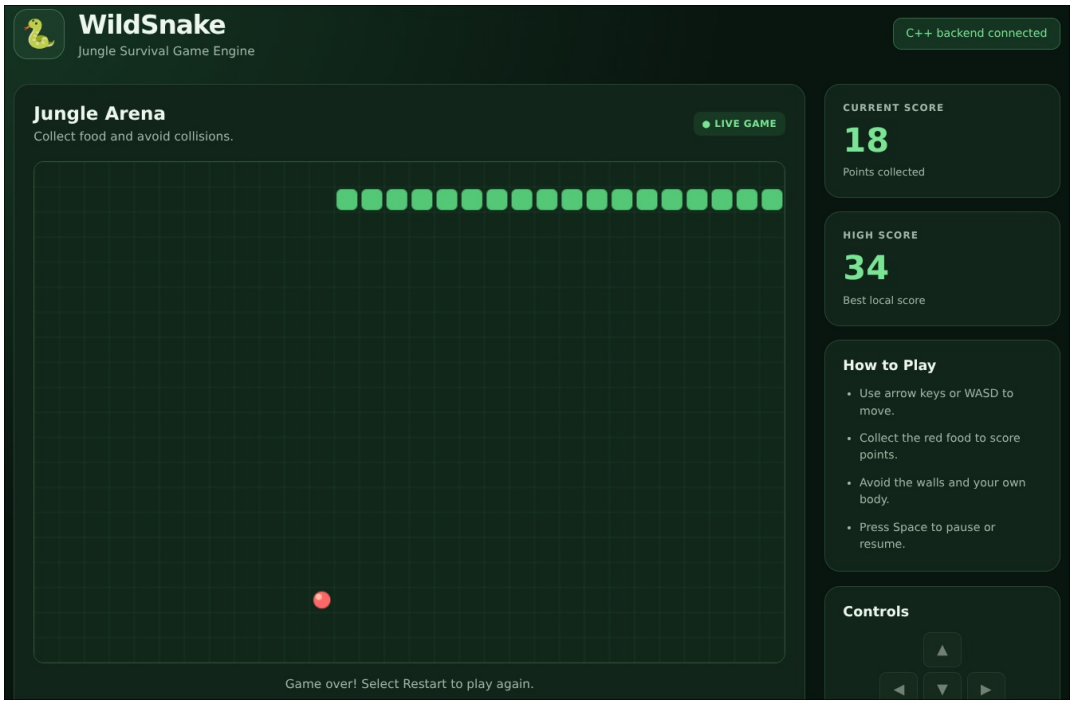


Figure 5.2: WildSnake game-over screen showing a score of 18 and a high score of 34

Chapter 6

Conclusion and Future Work

WildSnake is a useful OOP mini-project foundation: its engine is divided into small classes, exposes a browser API, and represents core gameplay concepts with private state and explicit interfaces. The TikZ diagrams describe the intended relationships and the current tick flow. The test source covers several basic cases, but the build and test outcome still needs to be verified.

Recommended next steps are to fix include/file-name consistency, implement or remove the currently undefined Grid template, connect persistent score management if high-score behavior is a requirement, and add tests for food, growth, collisions, and observer notifications. Further extensions could add configurable obstacles, difficulty levels, sound, and a packaged release.

References

1. ISO C++ Foundation, “C++ documentation,” https://isocpp.org/.
2. CMake, “CMake Documentation,” https://cmake.org/documentation/.
3. yhirose, “cpp-httplib,” https://github.com/yhirose/cpp-httplib.
4. nlohmann, “JSON for Modern C++,” https://github.com/nlohmann/json.
5. TikZ and PGF, “TikZ and PGF Manual,” https://tikz.dev/.
6. WildSnake project source code, supplied project repository, accessed [enter date].

Appendix A

Build and Run Guide

The CMake configuration declares the C++17 engine, HTTP server, smoke executable, and CTest target. From the project root, a typical build sequence is shown below. Dependency fetching may require network access at configure time. Run these commands after correcting the source naming and Grid issues; record the actual results in Chapter 6.

```
cmake -S backend -B build -DBUILD_TESTING=ON
cmake --build build
ctest --test-dir build --output-on-failure
```

Appendix B

Submission Completion Checklist

- Replace the institution name and repository access date if required by the department's format.
- Add genuine application screenshots in `report-assets/`.
- Fix the noted source/build issues and run the build and tests; report actual outcomes.
- Confirm the instructor's supplied template and apply any required cover-page, certificate, or formatting changes. The template mentioned in the request was not present in the workspace or chat attachment.
- Compile this file with a LaTeX installation that includes TikZ and listings, and inspect the generated PDF for layout and diagram issues.